

2018-Ph-H1-Q17

Section: Particles and Waves

Topic: Radiation

A radioactive sample has an activity of 800 Bq. Its half-life is 10 minutes.

What will its activity be after 30 minutes?

Solution:

30 minutes = 3 half-lives

$A = 800 \rightarrow 400 \rightarrow 200 \rightarrow 100 \text{ Bq}$

Answer: 100 Bq

Revision Tip:

After each half-life, divide activity by 2. Use the pattern:

$1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \dots$ based on how many half-lives have passed.